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Proposal to add the multiline option to `std::regex` for its ECMAScript engine

Introduction and Motivation

`std::regex` in the C++ standard has an engine that recognizes the ECMAScript grammar. But unlike the original `RegExp` of ECMAScript, this engine does not have a way to switch the behaviors of the regular expressions `^` and `$`, because there is not any option corresponding to the multiline property in ECMAScript. To fix this incompatibility and to make the behavior of this engine more compatible with the original `RegExp` of ECMAScript, I propose adding an option to `std::regex` for switching the behaviors of the regular expressions `^` and `$`.

Proposed Addition

1. To `syntax_option_type` [28.5.1]:

constexpr syntax_option_type multiline = unspecified ;

2. To Table 138 [28.5.1]:

Element: multiline

Effect(s) if set: Specifies that `^` shall match the beginning of every line and `$` shall match the end of every line, if the ECMAScript engine is specified.

Relevant Information

I reported this as a library issue once and it was listed as Issue #2503, but based on a recommendation I submit this matter as a proposal.